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Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech | ***Time: 03 Hours***

Code: 105601 | ***Subject: Compiler Design*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option for the following questions (Answer any seven): **[2 x 7 = 14]**

(a) Which phase of the compiler is responsible for removing comments and white spaces?
- (i) Syntax Analysis
- (ii) Lexical Analysis
- (iii) Semantic Analysis
- (iv) Intermediate Code Generation

(b) A lexical analyzer uses which of the following automata?
- (i) PDA
- (ii) Turing Machine
- (iii) DFA/NFA
- (iv) LBA

(c) LL(1) parsers use which parsing technique?
- (i) Bottom-up
- (ii) Shift-reduce
- (iii) Top-down
- (iv) Operator precedence

(d) Conflicts in LR parsing include
- (i) Shift-shift
- (ii) Shift-reduce
- (iii) Reduce-reduce
- (iv) Both b and c

(e) Type checking ensures
- (i) Syntax correctness
- (ii) Register allocation
- (iii) Semantic consistency
- (iv) Pipeline optimization

(f) Pipeline optimization helps reduce
- (i) Syntax errors
- (ii) Register conflicts
- (iii) Execution stalls
- (iv) Semantic errors

(g) Polymorphism is implemented using
- (i) NFA
- (ii) Heap
- (iii) Virtual functions
- (iv) Stack frames

(h) Which parser construction method is used by YACC?
- (i) LL
- (ii) LALR
- (iii) General LR
- (iv) SLR

(i) DAG stands for
- (i) Directed Applicable Graph
- (ii) Directed Acyclic Graph
- (iii) Data Analysis Graph
- (iv) Data Allocation Graph

(j) LEX tool is used for
- (i) Parser generation
- (ii) Lexical analysis
- (iii) Syntax analysis
- (iv) Intermediate code generation

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Q.2

(a) What are the various phases of a compiler? Explain each phase in detail by processing the input "a= (b+c)*(b+c)*2". [7]

(b) What is meant by lexical analysis, token, pattern and lexeme? Identify the lexemes that make up the token in the following program segment. Indicate the corresponding token and pattern. [7]

```
void swap(int i, int j)
{
    int t;
    t = i;
    i = j;
    j = t;
}
```

Q.3

(a) Construct set of LR(1) items for the given grammar and check whether it is SLR, CLR or LALR. [7]

```
S → CC
C → aC
C → d
```

(b) Consider the grammar [7]

```
E  → T E1,    E1 → +T E1 | ε
T  → F T1,    T1 → *F T1 | ε
F  → (E) | id
```

Find the FIRST and FOLLOW set of all the variables.

Q.4

(a) Consider the following grammar along with translation rules. [7]

```
S  → S1 # T    { S.val  = S1.val * T.val }
S  → T         { S.val  = T.val }
T  → T1 % R    { T.val  = T1.val + R.val }
T  → R         { T.val  = R.val }
R  → id        { R.val  = id.val }
```

Here # and % are operators and *id* is a token that represents an integer, and *id.val* represents the corresponding integer value. Using this translation scheme, compute the value of S.val for the root of the parse tree for the expression:

```
20 # 10 % 5 # 8 % 2 % 2
```

(b) Consider the below grammar, which defines simple arithmetic expressions. [7]

```
E → E + T | E - T | T
T → T * F | T / F | F
F → (E) | id
```

Now modify the above grammar with unconventional rules of precedence and associativity and then rewrite it such that,
the expression 3*2+1 will be interpreted as 3*(2+1).

the value of 1-2-3 will be 2 rather than -4.

Q.5

(a) Convert the regular expression (a|b)*ab into NFA, then convert the NFA into a minimal DFA. [7]

(b) Differentiate between synthesized and inherited attributes with example. [7]

Q.6

(a) What are shift-reduce and reduce-reduce conflicts? Explain with examples. [7]

(b) Differentiate between Top-down and Bottom-up parsing with example. [7]

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Q.7

(a) Consider the following C code segment: [7]

```
a = b + c;
e = a + 1;
d = b + c;
f = d + 1;
g = e + f;
```

In a compiler, this code segment is represented internally as a directed acyclic graph (DAG). Draw the DAG representation and find the number of nodes in the DAG.

(b) Explain architecture dependent code optimization techniques with examples. [7]

Q.8

(a) How are object-oriented features compiled in C++/Java-like languages? [7]

(b) Consider the following pseudo-code. [7]

```
L1: t1 = -1     L6: t6 = t5*M              L11: t3 = t3+1
L2: t2 = 0      L7: t7 = t4+t6             L12: if t3 < M goto L4
L3: t3 = 0      L8: t8 = a[t7]             L13: t2 = t2+1
L4: t4 = 4*t3   L9: if t8 <= max goto L11  L14: if t2 < N goto L3
L5: t5 = 4*t2   L10: t1 = t8               L15: max = t1
```

Find the number of basic blocks and the number of instructions in the largest basic block.

Q.9 Write short notes on any two of the following: **[7 x 2 = 14]**

(a) Symbol Table

(b) Peephole Optimization

(c) Activation Record

(d) Three Address Code

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